The Effect of Particle Size on Core-losses of Fe-Si-Cr Soft Magnetic Composites

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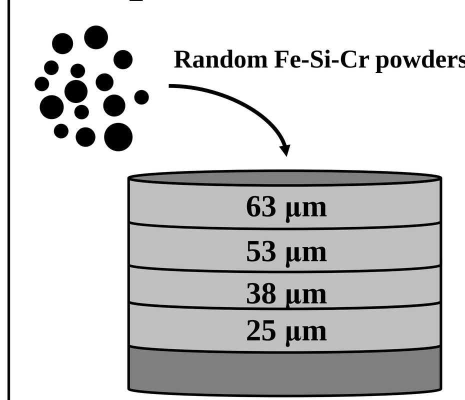
CPA-10

Introduction

❖Motivation

- ✓ Soft magnetic composites (SMCs) with high magnetic saturation (M_s), low coercivity (H_c), high inductance with high magnetic permeability and low core-loss are essential for high energy efficiency power-inductor at high frequencies.
  → iron-based soft magnetic alloy; Fe-Si-Cr alloy.
- ✓ Eddy-current loss reduction is essential for total core-loss reduction.
  → highly related with the particle size and resistivity of powders.
- ✓ We divided powder sizes of Fe-Si-Cr alloy into five parts by sieving the powder.
- ✓ We made toroidal core type samples depending on particle sizes, and measured permeability and core-loss to check if the core is suitable for high frequency application.

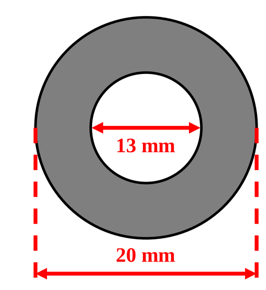
❖Experimental Process



Sieving

Divided into five powder sizes
S1: under 25 μm
S2: 25-38 μm
S3: 38-53 μm
S4: 53-63 μm
S5: over 63 μm

Mixed each powder with 3.0 wt% of epoxy resin.



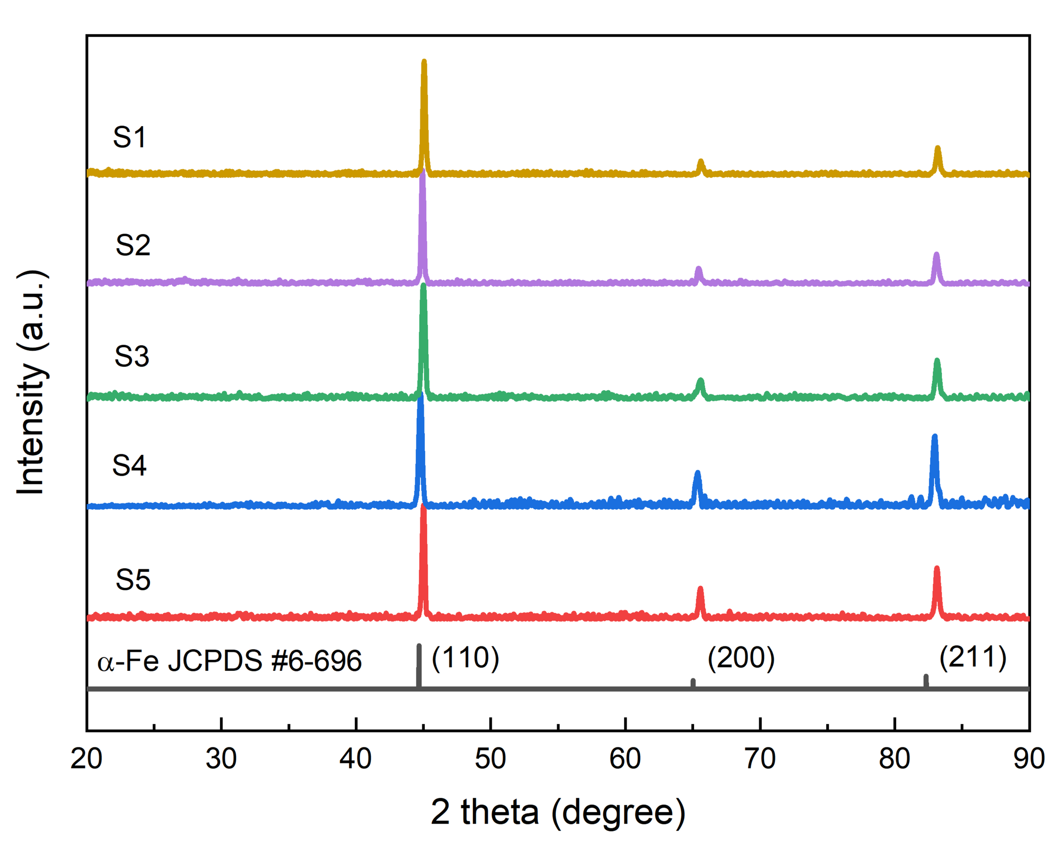
Fabrication of toroidal core

Dried the fabricated toroidal cores in 200 °C for two hours.

Take measurement of each sample: XRD, PSD, VSM, SEM/EDS, Impedance analyzer, B-H analyzer

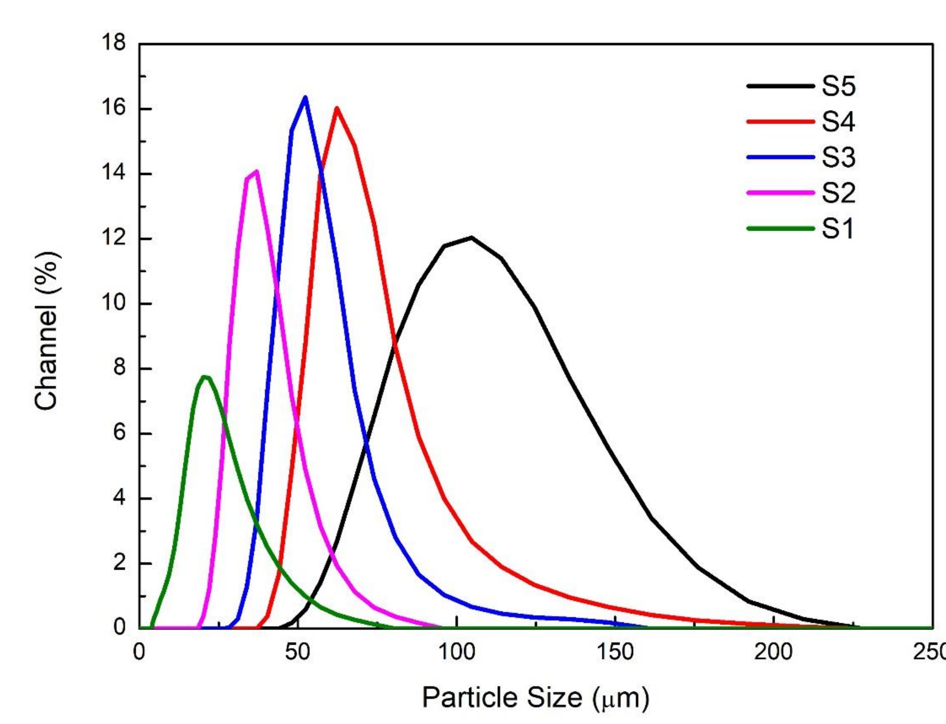
Experiments & Results

❖X-ray Diffraction (XRD)



- ✓ All powders were compared with simple cubic structure α-Fe (JCPDS #6-696) XRD peak.

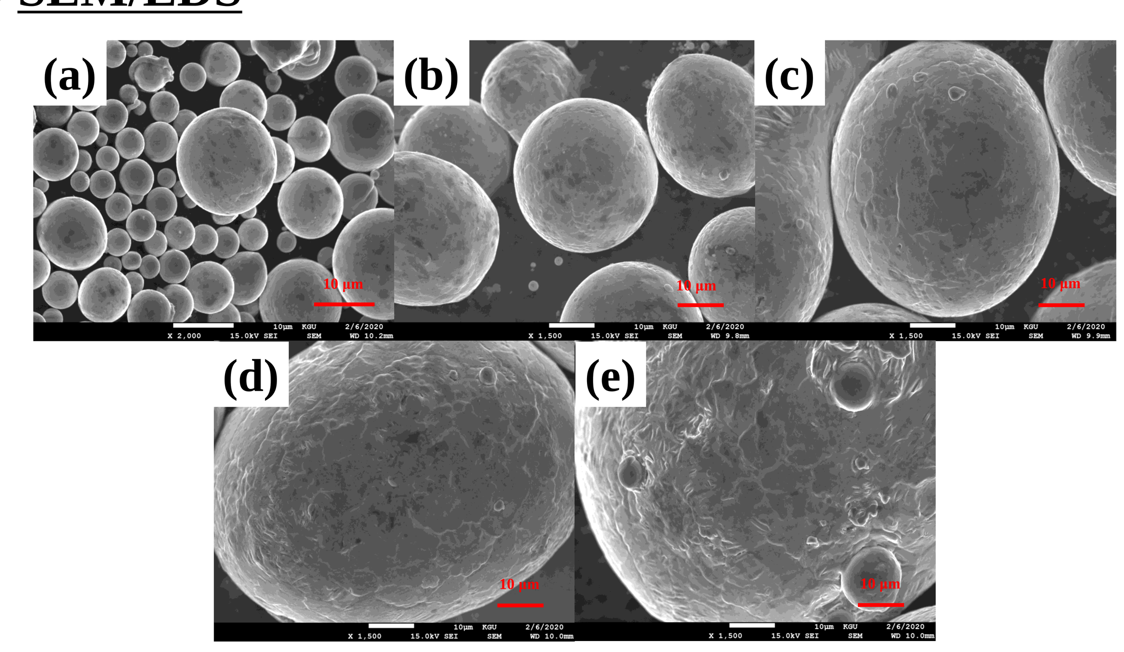
❖Particle Size Distribution (PSD)



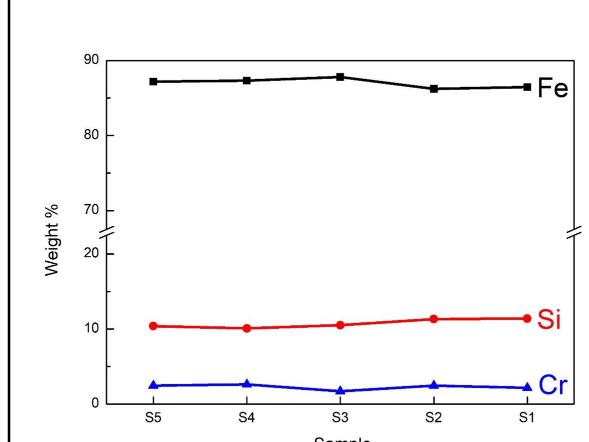
Sample	S1	S2	S3	S4	S5
D90 (μm)	35.31 μm	50.90 μm	70.64 μm	92.98 μm	140.05 μm
D50 (μm)	19.72 μm	35.23 μm	50.97 μm	63.89 μm	98.12 μm
D10 (μm)	10.03 μm	26.23 μm	39.32 μm	49.55 μm	68.52 μm

- ✓ S5 showed a very wide range of particle size distribution

❖SEM/EDS

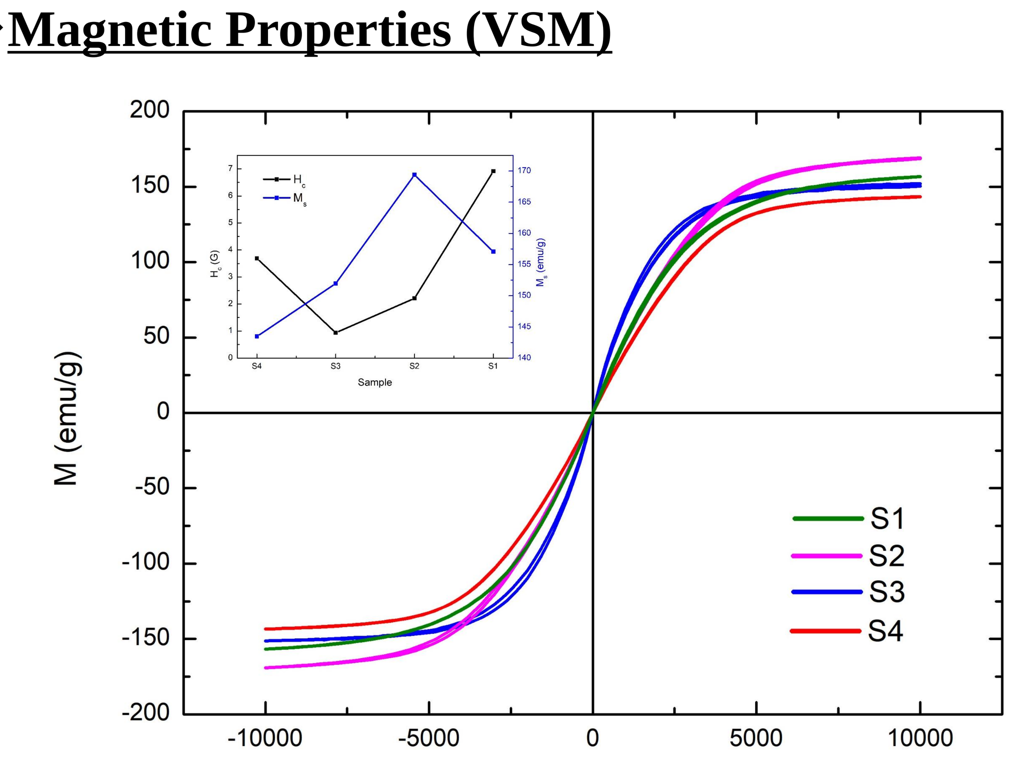


Morphology of Fe-Si-Cr powders (a) S1 (b) S2 (c) S3 (d) S4 (e) S5



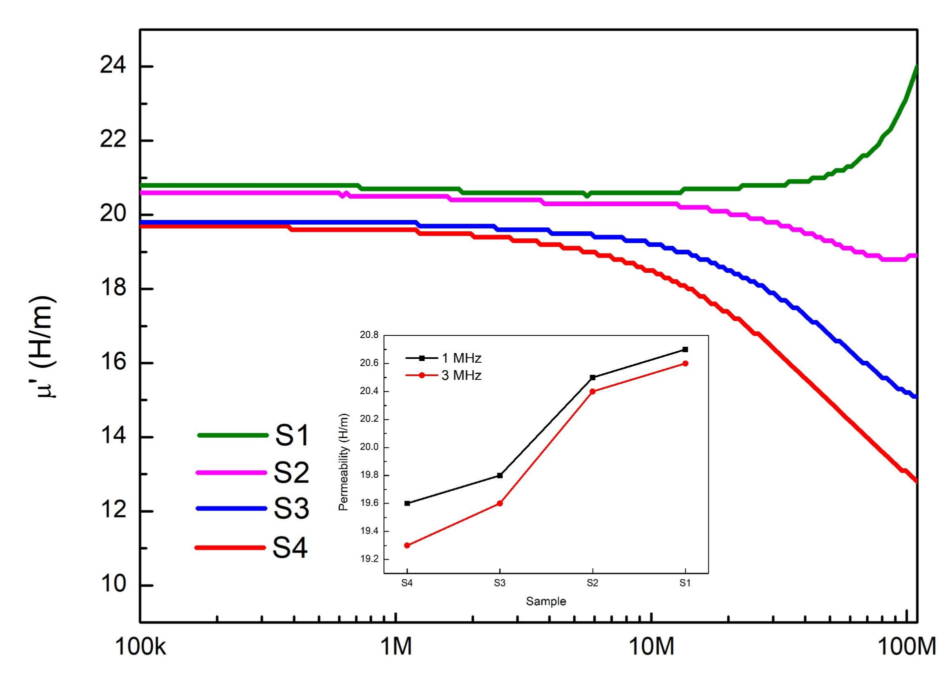
- ✓ All powders showed spherical shape because they were made by gas-atomization
- ✓ The average contents of Fe, Si, and Cr were about 87 wt%, 11 wt%, and 2 wt%, respectively.

❖Magnetic Properties (VSM)



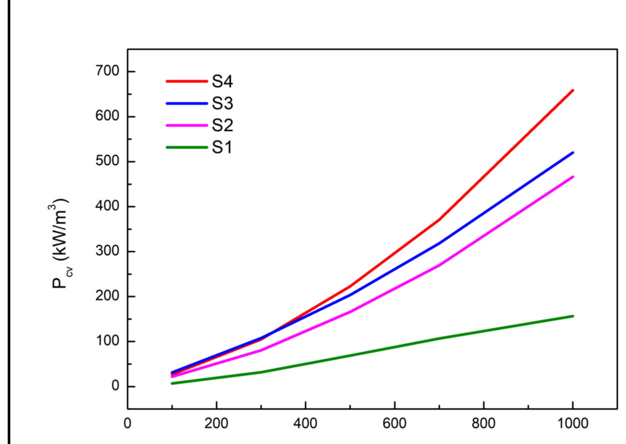
- ✓ All the powders showed good soft magnetic properties compared to conventional inductor materials such as ferrite-based materials, with high M_s, and low H_c.
- ✓ This good soft magnetic properties would give good impact on lowering hysteresis loss

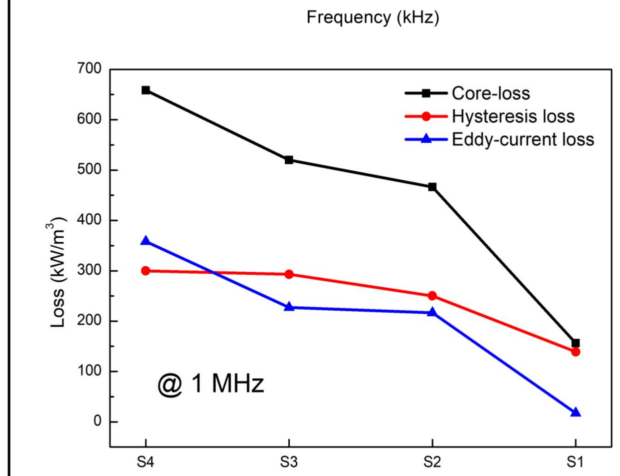
❖Permeability (Impedance Analyzer)



- ✓ Magnetic permeability of samples were measured by using impedance analyzer.
- ✓ The tendency showed that, as the particle size decreases, permeability increases.
- ✓ We mainly focused on the permeability at 1 MHz and 3 MHz, which are the frequencies usually used in various electronic devices.

❖Core-loss (B-H Analyzer)





- ✓ Total core-loss of each sample were measured by B-H analyzer.
- ✓ Samples with smaller particle sizes showed lower core-losses.
- ✓ Eddy-current and hysteresis losses were measured through curve fitting method.
- ✓ We could know that the main factor of core-loss reduction is eddy-current loss reduction.

Conclusions

- ✓ In this study, the crystal structure, particle size distribution, powder morphology, magnetic properties and core-losses of various cores with different particle sizes were investigated.
- ✓ The S1 sample with particle size of under 25 μm showed the highest permeability at both 1 MHz and 3 MHz. The tendency showed that as the particle size decreases, the permeability increases.
- ✓ Also, S1 showed the lowest total core-loss of 156.27 kW/m³, and we could know that the reduction of eddy-current loss was the main factor of core-loss reduction.
- ✓ It was expected that the smaller particle size can affect on the density and packing fraction of the fabricated core, which can be advantageous on improvement of permeability and lowering core-loss.